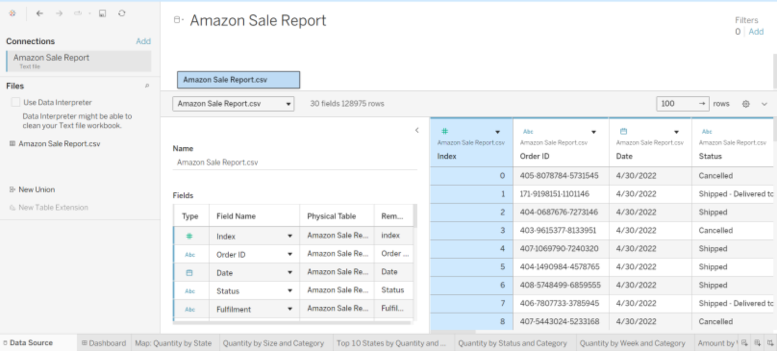
Project - Amazon Sales Dashboard

Introduction:
This Tableau research project analyzes Amazon sales data to uncover trends across geography, product size, shipping behavior, customer preferences, and order fulfillment. Using an interactive dashboard powered by pre-processed data from a .csv file, the project provides a comprehensive visual overview of quantities sold, sales amount, courier statuses, and shipping service levels. Through a blend of bar charts, maps, line graphs, donut charts, highlight tables, and KPIs, this dashboard supports stakeholders in understanding sales performance and guiding strategic decisions related to inventory, delivery logistics, and product focus.

Steps:

1. Load Data:

Loaded data by creating connection with 'Amazon Sales Report.csv' containing fields like Order ID, Date, Status, Asin, Category, Qty, Ship-State, Ship-Country, Qty, etc..



2. Sheet 1 - Map: Quantity by State

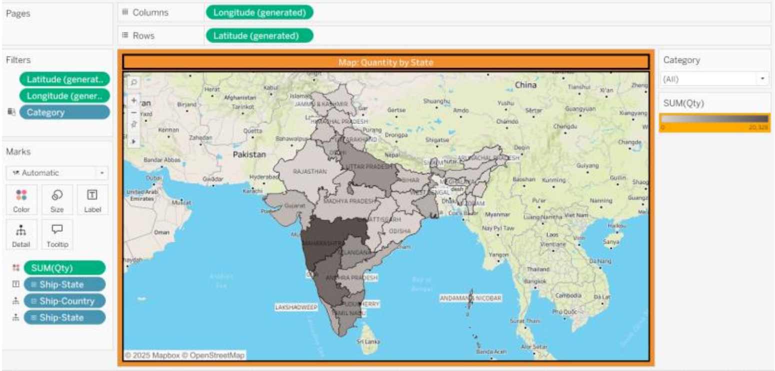
Objective: Show quantity sold per state using a filled map.

2.1. Dragged Fields to View:
- 2.1.1.Columns → Longitude (generated)
- 2.1.2.Rows → Latitude (generated)

2.2. Map Configuration:
- 2.2.1.Marks → Map
- 2.2.2.Dragged Ship-State and Ship-Country to Detail
- 2.2.3.Dragged Qty to Color
- 2.2.4.Dragged Ship-State to Label

2.3. Filters:
- 2.3.1. Dragged Category to Filters pane →Apply to Worksheets → All Using Related Data Sources

2.4. Formatting:
- 2.4.1.Used Color gradient (Gray Warm) to indicate quantity range
- 2.4.2.Formatted font size & color, background color, border color, etc.



3. Sheet 2: Quantity by Size and Category (Stacked Bar)

Objective: Compare quantity across sizes and categories in a stacked bar format.

3.1. Copied the previous sheet

3.2. Removed all Fields from View

3.3. Dragged Fields to View:

- 3.3.1. Columns → SUM(Qty)
- 3.3.2. Rows → Size

3.4. Marks Configuration:

- 3.4.1. Marks → Bar
- 3.4.2. Color → Dragged Category to Color to segment by Category
- 3.4.3. Label → Showed SUM(Qty) values on bars

3.5. Filters:

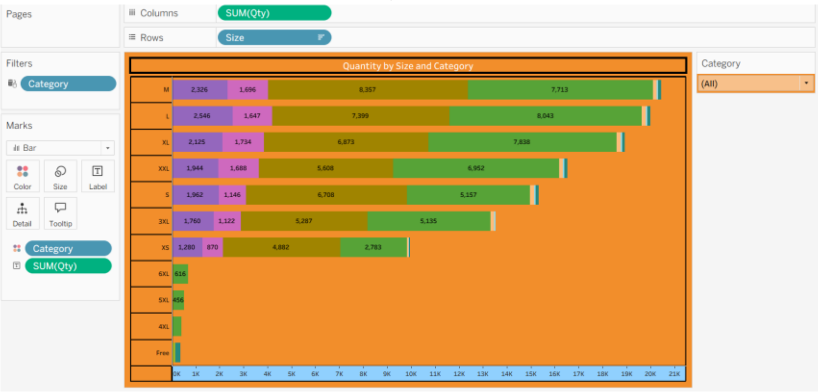
Kept Category to Filters pane for interactivity

3.6. Sort:

Sorted Size descending based on SUM(Qty) to show most sold sizes at top

3.7. Formatting:

Adjusted color scheme for different categories



4. Sheet 3: Top 10 States by Quantity & Category - Ship Service Level

Objective: Show top 10 shipping states and their distribution by shipping service level and category.

4.1. Copied the previous sheet

4.2. Removed all Fields from View

4.3. Dragged Fields to View:

- 4.3.1. Columns → SUM(Qty)
- 4.3.2. Rows → Ship-State

4.4. Created Ship Level Breakdown:

- 4.4.1. Dragged Ship-Service-Level to Rows
- 4.4.2. Marks → Bar

4.5. Color and Filters:

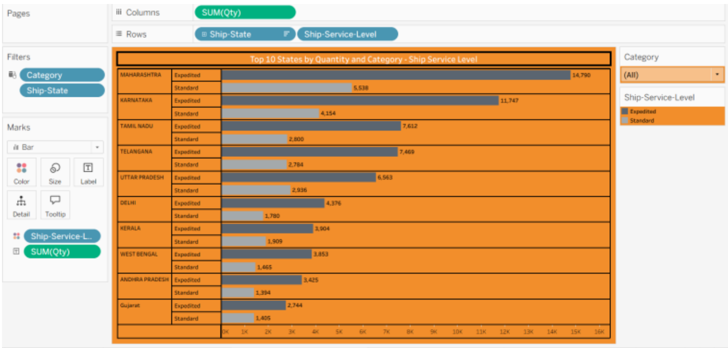
- 4.5.1. Dragged Ship-Service-Level to Color
- 4.5.2. Kept Category to Filters pane for interactivity
- 4.5.3. Added Ship-State to Filters for interactivity
- 4.5.4. Edited colors of Ship-Service-Level

4.6. Limit to Top 10 States:

Clicked Ship-State → Filter → Top → By Field → 10 by SUM(Qty)

4.7. Formatting:

- 4.7.1. Sorted by SUM(Qty) descending
- 4.7.2. Labeled each bar with quantity



5. Sheet 4: Quantity by Status and Category

Objective: Compare the quantity sold by order status (e.g., Shipped, Canceled) segmented by category to assess fulfillment performance.

5.1. Copied the previous sheet

5.2. Removed all Fields from View

5.3. Dragged Fields to View:

5.3.1. Columns → SUM(Qty)

5.3.2. Rows → Status

5.4. Color and Filters:

5.4.1. Kept Category to Filters pane for interactivity

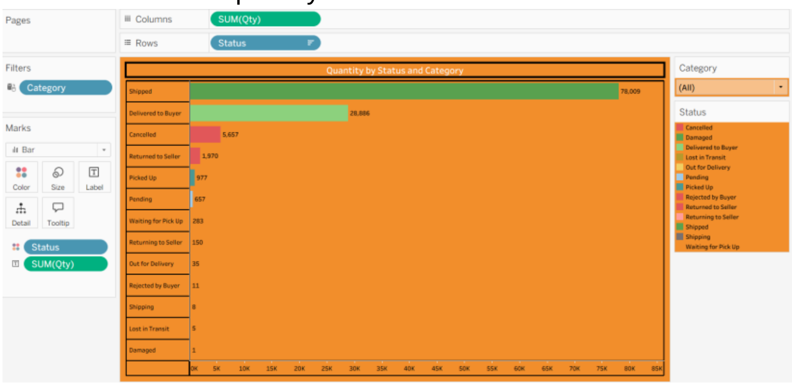
5.4.2. Dragged Status to Color

5.4.3. Edited colors of Status

5.5. Formatting:

5.5.1. Sorted by SUM(Qty) descending

5.5.2. Labeled each bar with quantity



6. Sheet 5: Quantity by Week and Category

Objective: Show weekly trends in product quantity sold, segmented by category to identify peak selling periods.

6.1. Copied the previous sheet

6.2. Removed all Fields from View

6.3. Dragged Fields to View:

6.3.1. Columns → Date (Week Number) → Week(Date)

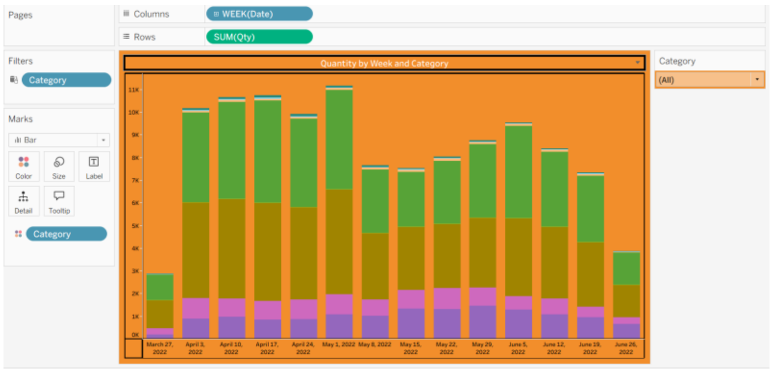
6.3.2. Rows → SUM(Qty)

6.4. Color and Filters:

6.4.1. Kept Category to Filters pane for interactivity

6.4.2. Dragged Category to Color

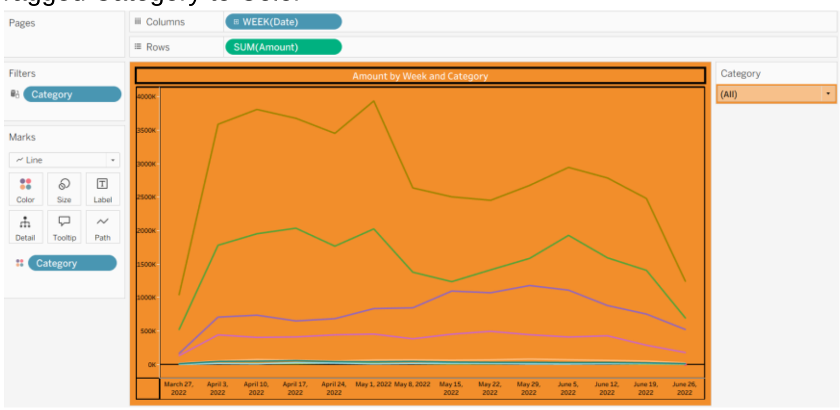
6.4.3. Edited colors of Category



7. Sheet 6: Amount by Week and Category

Objective: Analyze weekly revenue trends segmented by category to observe financial performance over time.

7.1. Copied the previous sheet
7.2. Removed all Fields from View
7.3. Dragged Fields to View:
  7.3.1. Columns → Date (Week Number) → Week (Date)
  7.3.2. Rows → SUM(Amount)
  7.3.3. Marks → Line
7.4. Color and Filters:
  7.4.1. Kept Category to Filters pane for interactivity
  7.4.2. Dragged Category to Color



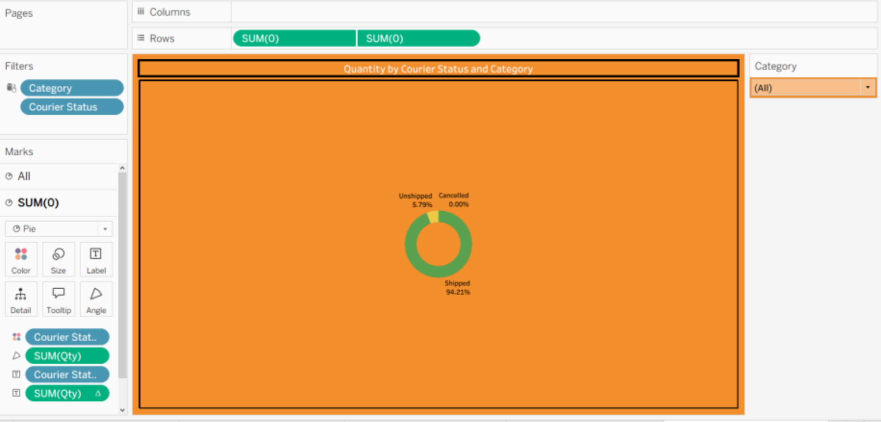
8. Sheet 7: Quantity by Courier Status and Category

Objective: Create a donut chart to visualize the distribution of quantities by courier status and category.

8.1. Copied the previous sheet
8.2. Removed all Fields from View
8.3. Dragged Fields to View:
  8.3.1. Rows → Double-click and enter 0 (to create a dummy axis)
  8.3.2. Marks → Pie
  8.3.3. Color → Courier Status
  8.3.4. Angle → SUM(Qty)
  8.3.5. Label → Courier Status, SUM(Qty)
8.4. Applied Quick Table Calculation:
  Clicked on SUM(Qty) in Label → Quick Table Calculation → Percent of Total
8.5. Color and Filters:
  8.5.1. Kept Category to Filters pane for interactivity
  8.5.2. Added Courier Status to Filters pane, removed Null from filter options
  8.5.3. Edited colors of Courier Status
  8.5.4. Hided Courier Status Filter
8.6. Created Dual-Axis Donut Shape:

8.6.1. Pressed Ctrl and duplicated the pie chart
8.6.2. On the second pie chart:
- Removed labels, color, and angle values
- Set color to Orange

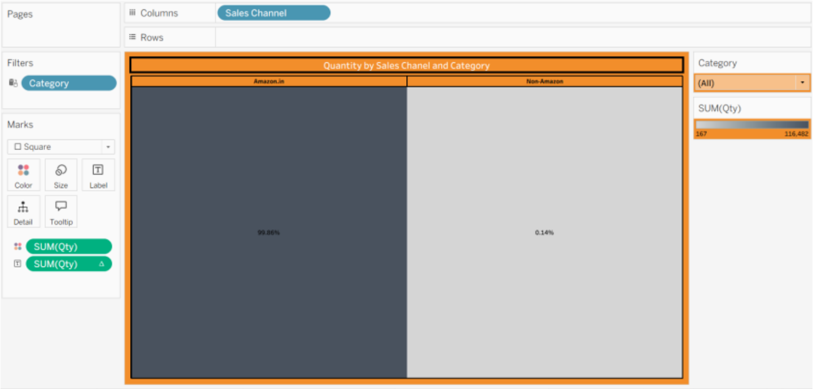
8.6.3. Increased Size for first pie chart
8.6.4. Right-clicked on the second axis → Selected Dual Axis



9. Sheet 8: Quantity by Sales Channel and Category

Objective: Visualize distribution of sales quantity across different sales channels segmented by product category.

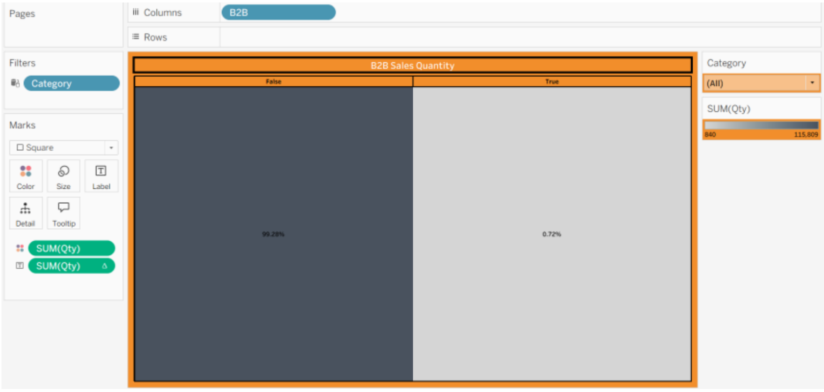
9.1. Copied the previous sheet
9.2. Removed all Fields from View
9.3. Dragged Fields to View:
9.3.1. Columns → Sales Channel
9.3.2. Marks → Square
9.3.3. Color → SUM(Qty)
9.3.4. Label → SUM(Qty)
9.4. Applied Quick Table Calculation:
Clicked on SUM(Qty) in Label → Quick Table Calculation → Percent of Total
9.5. Color and Filters:
9.5.1. Kept Category to Filters pane for interactivity
9.5.2. Edited SUM(Qty) Color to Gray



10. Sheet 9: B2B Sales Quantity

Objective: Isolate and display total sales quantity attributed to the B2B channel to support business customer analysis.

10.1. Copied the previous sheet
10.2. Removed Sales Channel from View
10.3. Dragged Fields to View:
Columns → B2B



11. Sheet 10: Currency

Objective: Display currency value used in the dataset, cleaned from null values.

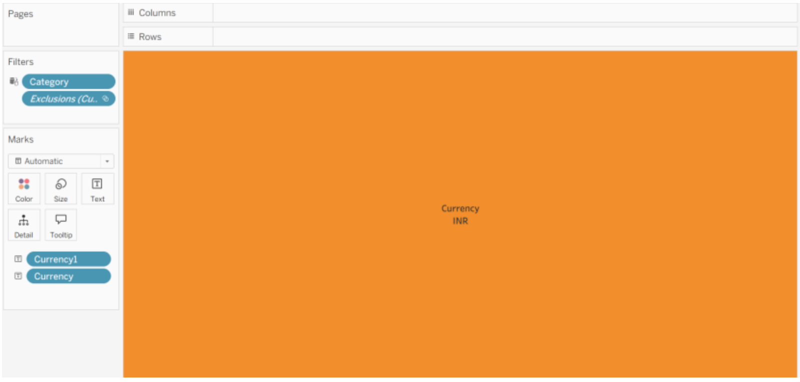
11.1. Copied the previous sheet

11.2. Removed All fields from View

11.3. Created Calculated field named ‘Currency1’

11.4. Dragged Currency1 and Currency to Text

11.5. Excluded Currency



12. Sheet 11: Total Quantity

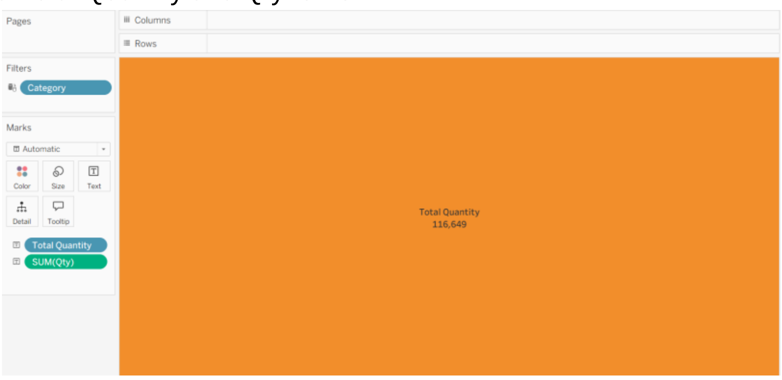
Objective: Display total quantity sold as a KPI using a calculated field.

12.1. Copied the previous sheet

12.2. Removed All fields from View

12.3. Created Calculated field named ‘Total Quantity’

12.4. Dragged Total Quantity and Qty to Text



13. Sheet 12: Total Amount

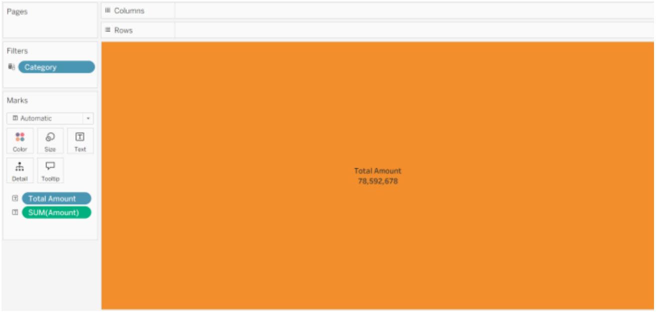
Objective: Display total sales amount as a KPI using a calculated field.

13.1. Copied the previous sheet

13.2. Removed All fields from View

13.3. Created Calculated field named ‘Total Amount’

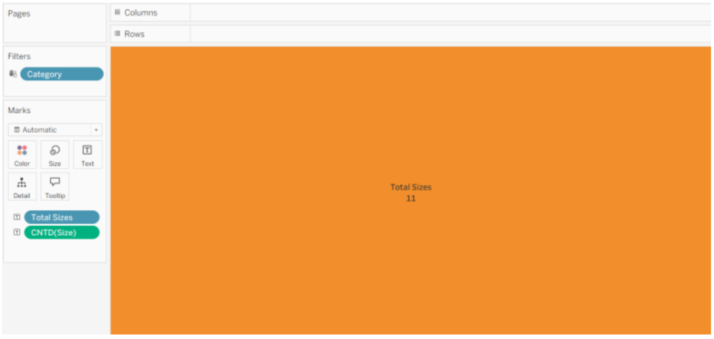
13.4. Dragged Total Amount and Amount to Text



14. Sheet 13: Total Sizes

Objective: Display the count of unique product sizes available in the dataset.

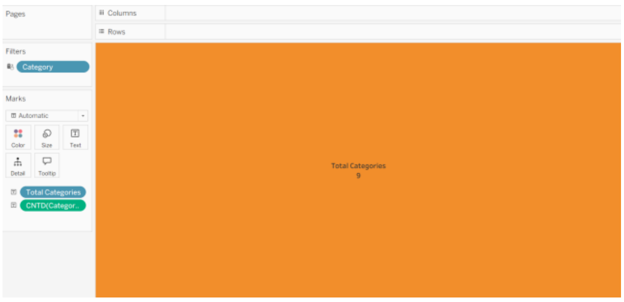
14.1. Copied the previous sheet
14.2. Removed All fields from View
14.3. Created Calculated field named 'Total Sizes'
14.4. Dragged Total Sizes and Size to Text
14.5. Set Size Measure to Count(Distinct)



15. Sheet 14: Total Categories

Objective: Display the count of unique product categories.

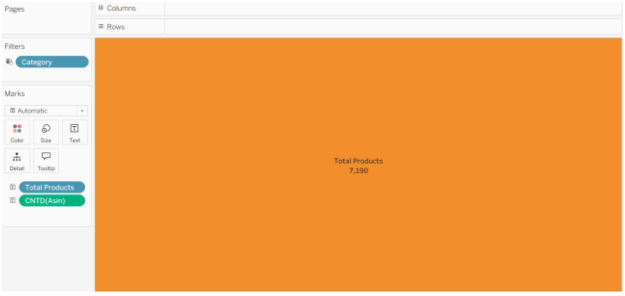
15.1. Copied the previous sheet
15.2. Removed All fields from View
15.3. Created Calculated field named 'Total Categories'
15.4. Dragged Total Categories and Category to Text
15.5. Set Category Measure to Count(Distinct)



16. Sheet 15: Total Products

Objective: Display the number of distinct ASINs (product identifiers) to show catalog diversity.

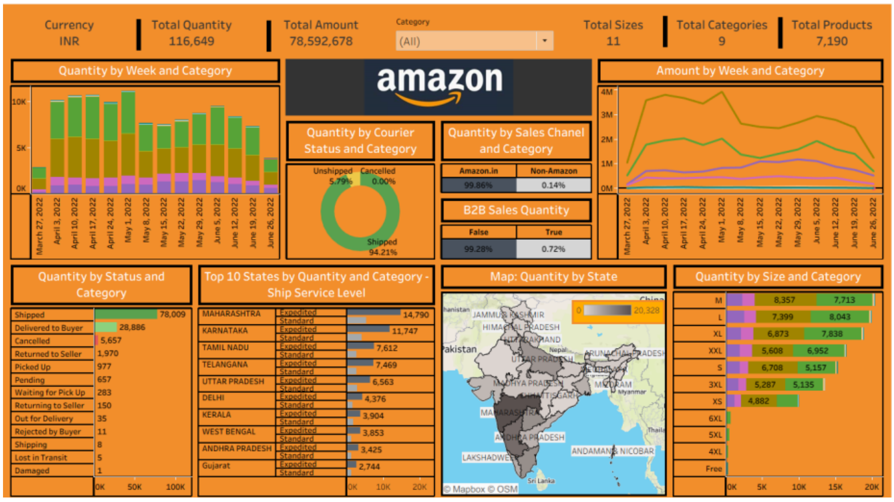
16.1. Copied the previous sheet
16.2. Removed All fields from View
16.3. Created Calculated field named 'Total Products'
16.4. Dragged Total Categories and Asin to Text
16.5. Set Asin Measure to Count(Distinct)



17. Dashboard

17.1. Placed all the sheets to the dashboard.

17.2. Added logo of Amazon.



Key Findings:

- **Peak Periods:** For **Quantity by Week**, there appear to be noticeable peaks around the 1st week of April, 2022 to May, 2022, with quantities reaching close to the 10,000 mark. For **Amount**, similar peaks are observed within this time period, where the amount reaches near the 4,000,000 INR mark.
- **High Shipment Success Rate:** A significant **94.21%** of the quantity has been shipped, with only **5.79%** unshipped and **0.00%** cancelled.
- **Dominant Sales Channel:** Amazon.in accounts for an overwhelming **99.86%** of the total quantity sold, while non-Amazon channels constitute only **0.14%**. This indicates a near-exclusive reliance on Amazon for sales.
- **Most Shipments are B2C: 99.39%** of the quantity represents B2C (Business-to-Consumer) sales, with a mere **0.61%** being B2B (Business-to-Business) sales.
- **High Return to Seller Rate:** A notable **1,970 units** have been returned to the seller, which is a substantial figure compared to other return/cancellation statuses.
- **Top Contributing States: Maharashtra** (14,790 units), **Karnataka** (11,747 units), and **Tamil Nadu** (7,612 units) are the top three states in terms of quantity shipped.
- **Top Performing Sizes:** Sizes **M** is the highest selling size by quantity.
- **Overall Metrics:** The total quantity sold is **116,649** units, amounting to **78,592,678 INR**. There are **11** total sizes, **9** total categories, and **7,190** total products.

Conclusion:

The Amazon Sales Dashboard created in Tableau presents a holistic and interactive visualization of e-commerce sales performance. By integrating quantity, revenue, shipping, product size, and regional data into visual formats, the dashboard empowers users to explore multi-dimensional insights with ease. The project effectively identifies high-performing states, peak sales weeks, popular sizes, and fulfillment trends. Additionally, KPIs offer quick snapshots of business scale - such as total quantity, revenue, product diversity, and category range. This interactive solution enables data-driven decision-making and offers a scalable foundation for advanced analytics such as forecasting, churn detection, or customer segmentation.